

C023115

Software Testing- Unit 5

Levels of Testing(10M)



System Testing

- System Testing is a type of software testing that evaluates the overall functionality and performance of a complete and fully integrated software solution.
- It tests if the system meets the specified requirements and if it is suitable for delivery to the end-users.
- This type of testing is performed after the integration testing and before the acceptance testing.
- This stage checks whether the entire system functions as expected in a real-world environment.
- It includes both functional and non-functional tests to ensure that the software meets customer needs.
- Full end-to-end testing of the software.
- Verifying both functional and non-functional requirements.
- Testing the software's behavior in real-world conditions.

Recovery Testing

- This type of testing evaluates how well a software system can recover from various failures, such as crashes, hardware failures, or network outages.
- It ensures that the system can gracefully restore data, resume operations, and maintain data integrity after encountering an issue.

Security Testing

- This focuses on identifying vulnerabilities within the software system and protecting it from unauthorized access, data breaches, and other security threats.
- It involves verifying security mechanisms, conducting penetration tests, and ensuring compliance with security standards.

Performance Testing

- This assesses the system's responsiveness, stability, and scalability under various workloads.
- It measures metrics like speed, throughput, resource utilization, and response times to ensure the system meets performance benchmarks.

Stress Testing

- A sub-type of performance testing, stress testing evaluates the system's behavior under extreme workloads, pushing it beyond its normal operational limits to identify its breaking point and how it handles overload conditions.

Usability Testing

- This focuses on evaluating the ease of use and user-friendliness of the software.
- It involves observing users interacting with the system to identify any usability issues, assess the user experience, and ensure the interface is intuitive and efficient.

Compatibility Testing

- This verifies that the software functions correctly across different environments, including various operating systems, browsers, hardware configurations, and network conditions. It ensures consistent behavior and functionality regardless of the platform.

Acceptance Testing

- Acceptance Testing is a formal testing according to user needs, requirements, and business processes conducted to determine whether a system satisfies the acceptance criteria or not and to enable the users, customers, or other authorized entities to determine whether to accept the system or not.
- Acceptance Testing is the last phase of software testing performed after System Testing and before making the system available for actual use.

Acceptance Criteria

- Acceptance criteria are essential components in project management and software development, serving as a clear set of conditions that a product or feature must meet to be considered complete and acceptable by the stakeholders.
- These criteria help ensure that everyone involved has a shared understanding of what "done" looks like, reducing ambiguity and aligning expectations.

Types of acceptance criteria:

Functional: Describes the desired functionality, behavior, or features of the software.

Non-Functional: Focuses on aspects such as performance, security, usability, and scalability.

Positive: Describes the expected positive outcomes when the feature works correctly.

Negative: Outlines the behavior or outcomes when the feature fails or encounters errors.

Alpha Testing

- Alpha Testing is performed by internal teams to catch major bugs early in development.
- Alpha testing is a crucial software testing phase, typically conducted by the development team before releasing the software for Beta Testing.
- It is an internal process designed to identify bugs and defects in the system. In this phase, developers and testers focus on finding critical issues within the application.

Who is Involved in Alpha Testing?

Developers, testers, and QA teams.

Beta Testing

- Beta Testing involves a select group of external users who test the software in real-world conditions, offering valuable feedback.
- Beta Testing is the phase that follows Alpha Testing, where the software is made available to a selected group of external users outside the organization.
- The primary objective of Beta Testing is to uncover any remaining bugs, performance issues, or usability concerns that were not identified during Alpha Testing.
- **Who is Involved in Beta Testing?**
External users, developers, and product managers.

Smoke Testing

- Commonly known as build verification testing, smoke testing is a functional testing type that determines whether the deployed build is stable or not.
- If the build is stable, the QA team can continue with other types of testing.
- Smoke testing identifies issues in the early stages before testing starts and saves a lot of time and effort.

When should you run smoke tests?

- Smoke tests are run when new features are added to the build. If the new build passes these tests, the testing process can continue.

Sanity Testing

- Sanity testing is defined as a subset of regression testing and takes place after it. Once all proposed code fixes are completed, sanity testing is done to ensure smooth performance and that the fixes ensure the products work as expected. .

Sanity Testing

- Suppose we have launched a clothing store application. A few days after our launch, users report that after adding items to the cart, the ‘Continue Shopping’ option is not working. This bug is taken up by the development team and they fix the code. This bug fix is then taken up by the testing team to run a sanity test. After a green flag from the testing team, a new build is triggered.

Sanity Testing

- The main advantage of sanity testing is that it is a quick method since there are no predefined test cases or documentation. It is helpful in cases where bug fixes are required on an urgent basis.

When should you run sanity tests?

- After regression testing is done and minor changes are introduced in the code. It is done before the build goes into production.

Accessibility Testing

- Accessibility testing is the practice of making your web and mobile apps usable to as many people as possible.
- It makes apps accessible to those with disabilities, such as vision impairment, hearing disabilities, and other physical or cognitive conditions.
- This testing verifies that the software complies with accessibility standards like WCAG (Web Content Accessibility Guidelines) and is inclusive for all users.